EX NO: 9

PERFORMANCE EVALUATION OF AM GENERATION AND DETECTION UNDER NOISY CHANNEL CONDITIONS USING MATLAB

PROGRAM (MATLAB CODE):

```
clc;
clear;
close all;
% Parameters
fs = 100000;
t = 0:1/fs:0.02;
fm = 500;
fc = 5000;
% Message Signal
m = cos(2*pi*fm*t);
c = cos(2*pi*fc*t);
% AM Modulation
am = (1 + m).*cos(2*pi*fc*t);
% Add AWGN Noise
noisy = awgn(am,10,'measured'); % 10 dB SNR
% Envelope Detection
env = abs(hilbert(noisy));
% Display Results
figure;
subplot(5,1,1)
plot(t,m)
title('Message Signal')
grid on
subplot(5,1,2)
plot(t,c)
```

```
title('Carrier Signal')
grid on
subplot(5,1,3)
plot(t,am)
title('AM Signal')
grid on
subplot(5,1,4)
plot(t,noisy)
title('AM Signal with AWGN Noise')
grid on
subplot(5,1,5)
plot(t,env)
title('Recovered Signal')
grid on
```

OUTPUT:

